

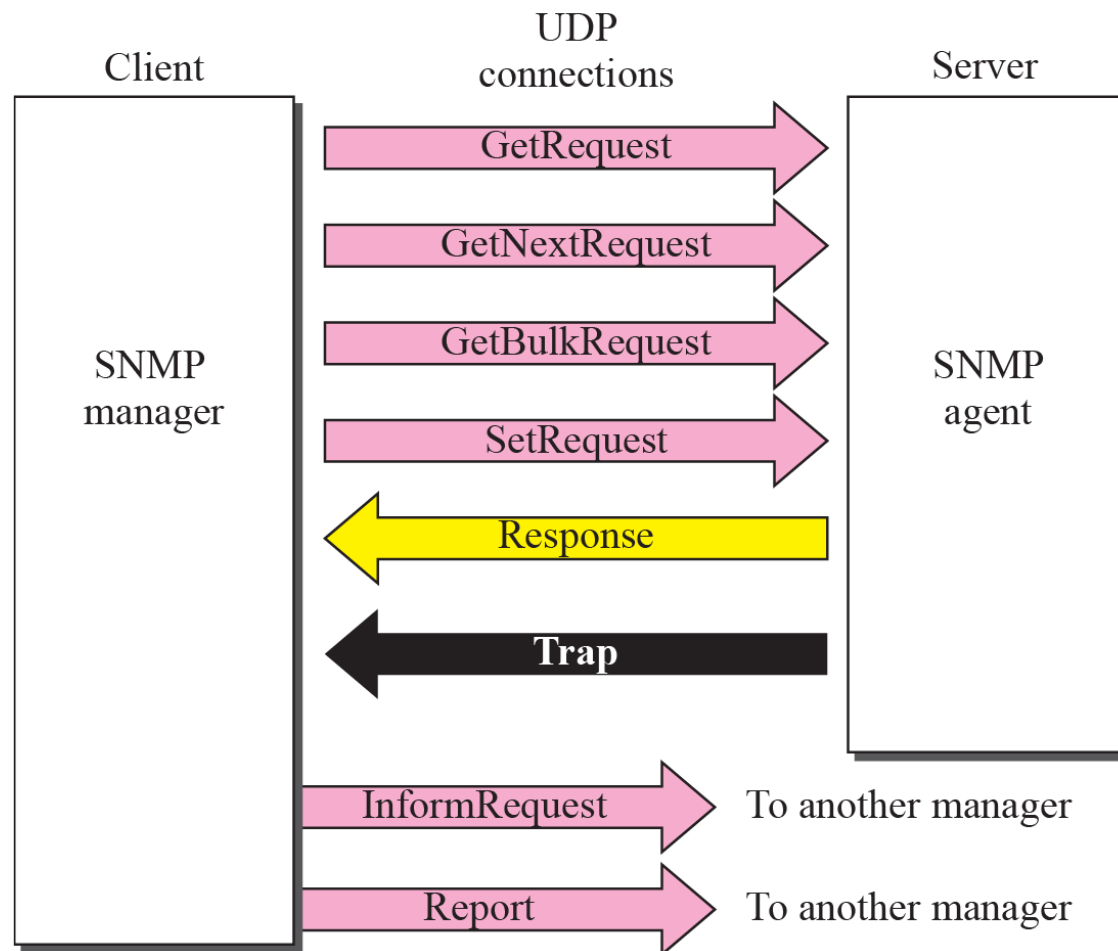
GESTÃO DE REDES / NETWORK MANAGEMENT
Notas complementares / Complementary notes

SNMP Protocol Data Units

Basic Encoding Rules



SNMP PDUs



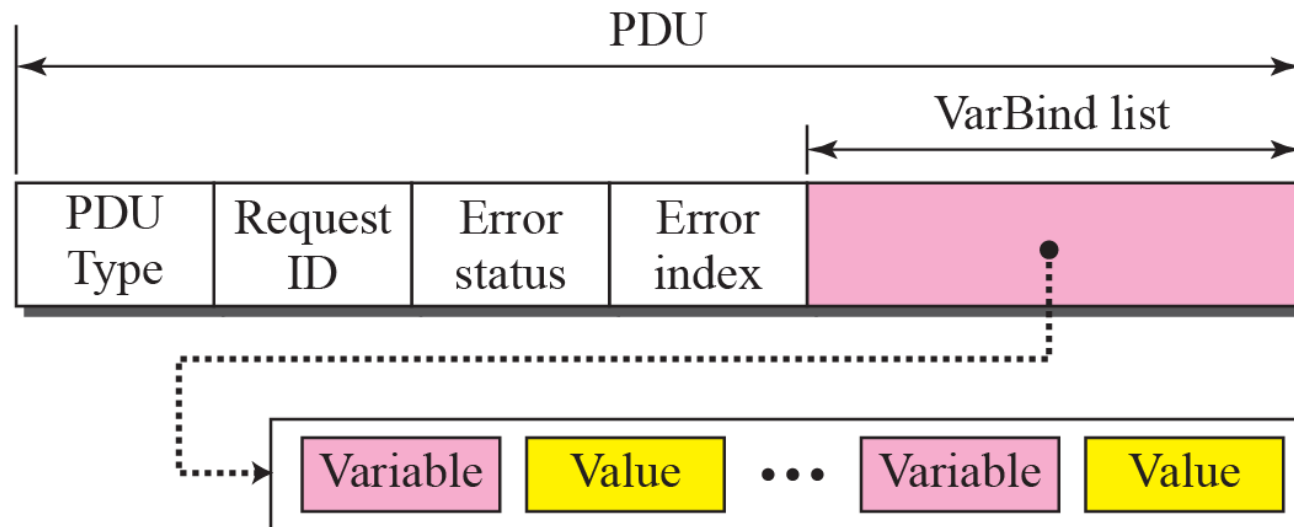
*Livro "TCP/IP Protocol Suite",
Behrouz A. Forouzan, McGraw-Hill (capítulo 24)*



SNMP Messages (v1/v2c)

```
SNMPv1/SNMPv2c Message {  
    version      snmpv1=0 snmpv2c=1  
    community   STRING  
    PDU (operation) {  
        request-id          INTEGER  
        error-status        INTEGER  
        error-index         INTEGER  
        variable-bindings {  
            OID, VALUE  
            ...  
        }  
    }  
}
```

SNMP PDUs



Differences:

1. Error status and error index values are zeros for all request messages except GetBulkRequest.
2. Error status field is replaced by non-repeater field and error index field is replaced by max-repetitions field in GetBulkRequest.

SNMP PDUs



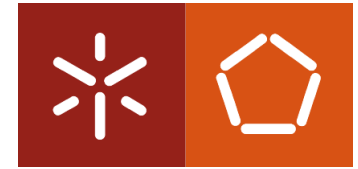
Table 24.3 *PDU Types*

Type	Tag (Binary)	Tag (Hex)
GetRequest	10100000	A0
GetNextRequest	10100001	A1
Response	10100010	A2
SetRequest	10100011	A3
GetBulkRequest	10100101	A5
InformRequest	10100110	A6
Trap (SNMPv2)	10100111	A7
Report	10101000	A8

Table 24.4 *Types of Errors*

Status	Name	Meaning
0	noError	No error
1	tooBig	Response too big to fit in one message
2	noSuchName	Variable does not exist
3	badValue	The value to be stored is invalid
4	readOnly	The value cannot be modified
5	genErr	Other errors

SNMP Messages (v3)



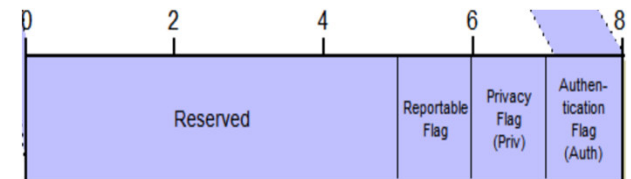
```
SNMPv3Message {  
  Version  
  
  HeaderData {  
    msgID  
    MaxSize  
    Flags  
    SecurityModel  
  
  }  
  UsmSecurityParameters STRING {  
    AuthoritativeEngineID  
    AuthoritativeEngineBoots  
    AuthoritativeEngineTime  
    UserName  
    AuthenticationParameters  
    PrivacyParameters  
  
  }  
  ScopedPduData {  
    contextEngineID  
    contextName  
    PDU  
  
  }  
}
```

INTEGER
(snmpv3 = 3)

INTEGER
INTEGER
STRING
INTEGER
(USM = 3)

STRING,
INTEGER
INTEGER
STRING
STRING
STRING

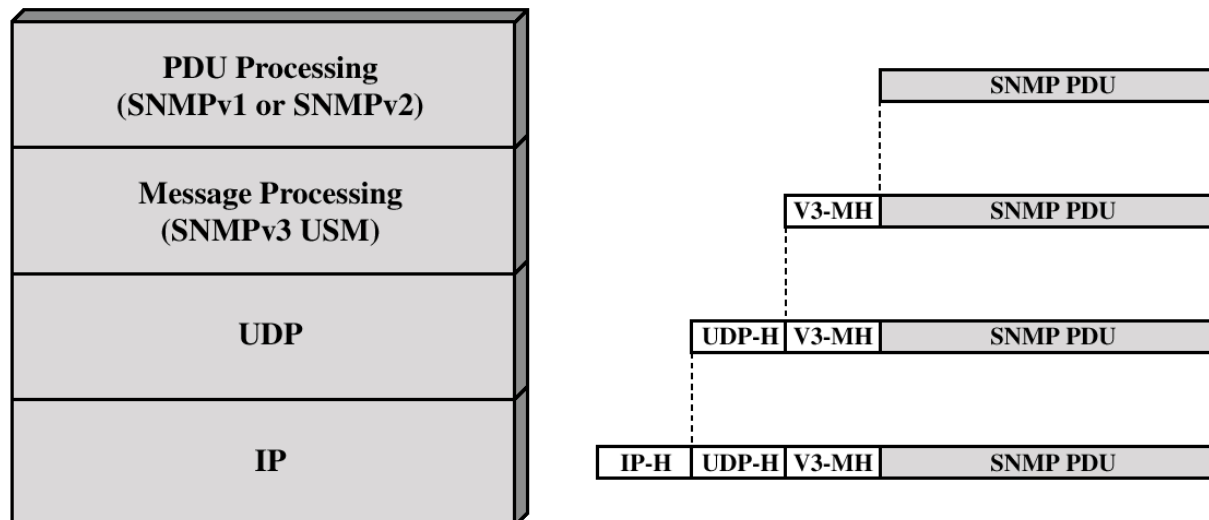
STRING,
STRING,
SNMPv2 PDUs





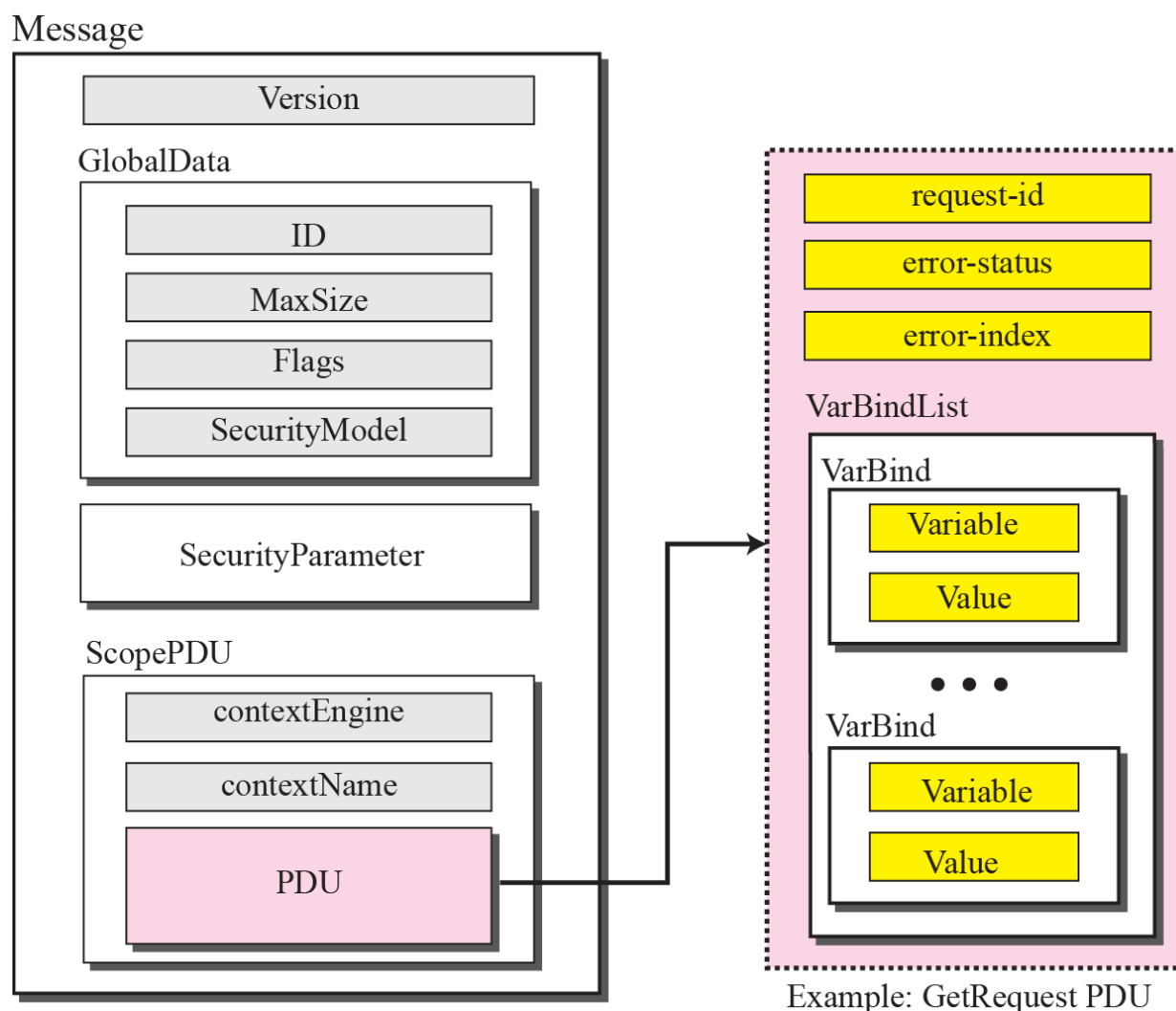
SNMP Messages (v3)

- **SNMPv3 defines a new message format that includes security and encapsulates v1/v2c messages**

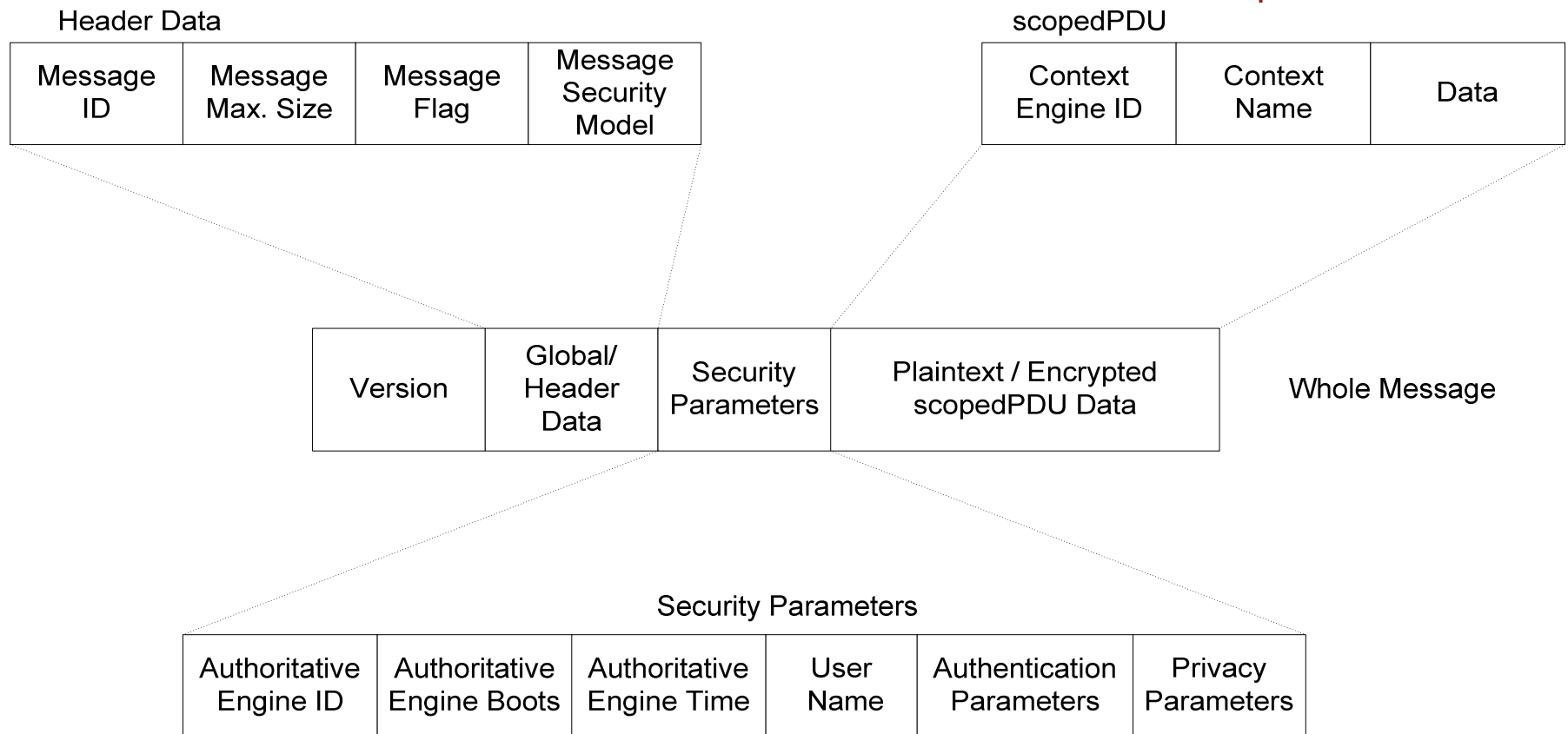


IP-H = IP header
UDP-H = UDP header
V3-MH = SNMPv3 message header
PDU = Protocol data unit

SNMP Messages (v3)



SNMP Messages (v3)



SNMPv3 Message Format

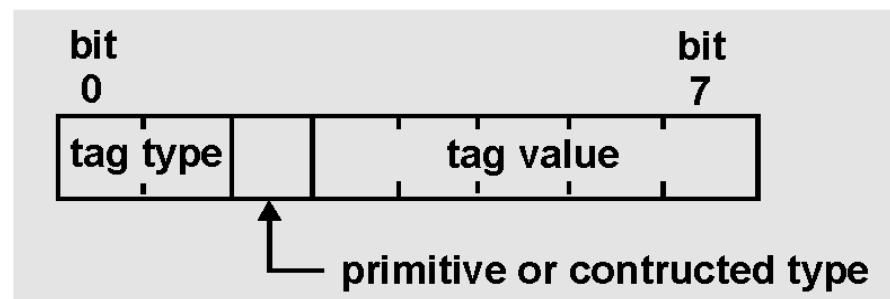


BER – Basic Encoding Rules

- **The data types defined in ASN.1 must be sent between heterogeneous systems**
 - We need a universal encoding scheme, independent of the hardware/operating system/software (encoding rules)
- **BER: Basic Encoding Rules**
 - TLV Triplets (Tag, Length, Value) for each type:



- 8 bit Tags:



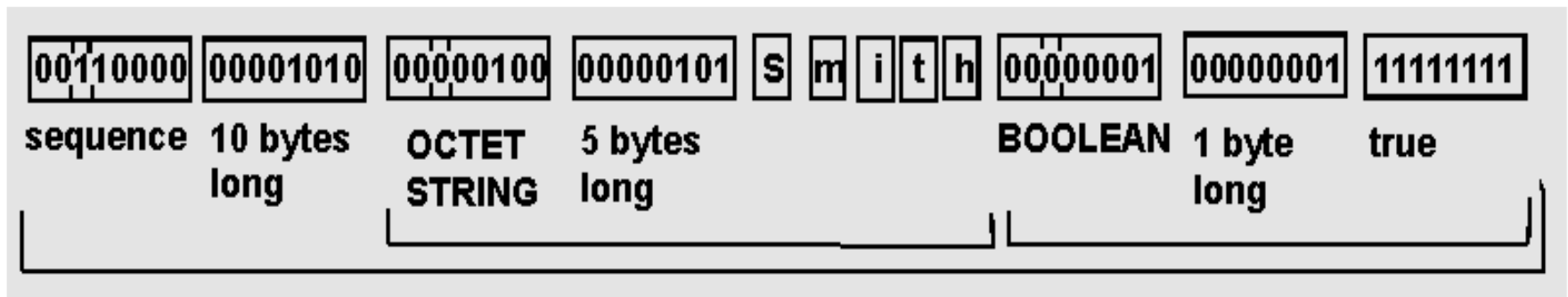


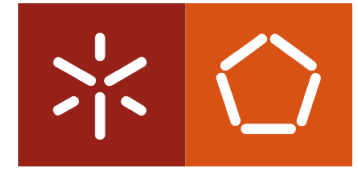
BER – Basic Encoding Rules

- Example for the following ASN.1 Sequence:

```
Attendee ::= SEQUENCE {  
    name  OCTET STRING,  
    paid   BOOLEAN  
}
```

- The BER byte stream for the record data **“Smith”, T** is:

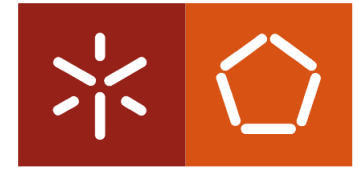




BER – Basic Encoding Rules

- **BER is defined in ITU-T standard X.690**
- **X.690 defines subsets of BER (simplified versions, with pragmatic decisions, no choices):**
 - Canonical Encoding Rules (CER)
 - Distinguished Encoding Rules (DER)
- **But there are others available:**
 - XML Encoding Rules (XEL)
 - JSON Encoding Rules (JER)
 - ...
- **Question/discussion: How efficient is BER encoding?**
 - ... ?

BER – Basic Encoding Rules



- **ASN.1 compilers generate definitions for programming languages with:**
 - Data structures, and
 - Functions for Encoding/Decoding those data structures

